

Proof by Contraposition

- We wish to prove implication $P(c) \rightarrow Q(c)$ is true for an arbitrary c in the domain
- Recall that

$$\underline{P(c) \rightarrow Q(c)} \equiv \underline{\neg Q(c) \rightarrow \neg P(c)}$$

- We can prove $P(c) \rightarrow Q(c)$ by proving $\neg Q(c) \rightarrow \neg P(c)$
- **Proof strategy:** To prove $P(c) \rightarrow Q(c)$, we need to show that if conclusion $Q(c)$ is F, then $P(c)$ is false too

Example

Prove that for any integer n if n^2 is odd, then n is odd.

$$\neg Q \rightarrow \neg P$$

if n is even, then n^2 is even

Assume n is even.

$$\textcircled{n = 2k}, \quad \underline{k \in \mathbb{Z}}$$

$$n^2 = (2k)^2 = 4k^2 = \textcircled{2} \textcircled{\underline{(2k^2)}} \rightarrow \text{integer}$$

Since $2k^2$ is an integer, n^2 is even □

def:
 $n \text{ even} \rightarrow 2k$
 $n \text{ odd} \rightarrow 2k+1$

Observation

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Combining these two, we can write:

For any integer n , n^2 is odd *if and only if* n is odd.

Example

P

Q

For integers m and n , show that if nm is even, then either m or n is even.

$\neg Q \rightarrow \neg P \equiv$ if m and n are both odd then
 nm is odd

Assume
that m and n are odd. , $k, l \in \mathbb{Z}$

$$m = 2k + 1, \quad n = 2l + 1$$

$$\begin{aligned} nm &= (2k+1)(2l+1) \\ &= 4kl + 2k + 2l + \underline{1} \\ &= 2(2kl + k + l) + 1 \end{aligned}$$

Since $2kl + k + l$
is an integer.
~~then~~ and mn
has a $2t + 1$ format
 $\rightarrow mn$ is odd $\square$

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- **Proof strategy:** To prove $P(c) \rightarrow Q(c)$, we show that $P(c) \wedge \neg Q(c)$ leads to a *contradiction*, i.e., a false proposition.
That is, $P(c) \wedge \neg Q(c) \equiv \text{F}$, or

$$P(c) \wedge \neg Q(c) \rightarrow \text{F}$$

must be true.

Example

P Q

~~✱~~ Prove that for any integer n if n^2 is odd, then n is odd.

$P \wedge \neg Q \equiv n^2 \text{ is odd and } n \text{ is even}$

$$n = 2k, \quad k \in \mathbb{Z}$$

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

$\rightarrow n^2$ is even

We assumed n^2 is odd, but we found n^2 is even!

This is a contradiction.

□

Proof by cases

- We wish to prove $P(c) \rightarrow Q(c)$ is true for any c in domain
- Suppose the proposition $P(c)$ can be expressed in several cases:

$$P_1(c) \vee P_2(c) \vee P_3(c) \vee \cdots \vee P_n(c)$$

- We need to show that the following is true:

$$\neg \left(P_1(c) \vee P_2(c) \vee P_3(c) \vee \cdots \vee P_n(c) \right) \vee Q(c)$$

- Or equivalently, the following is true

$$\left(\neg P_1(c) \vee Q(c) \right) \wedge \left(\neg P_2(c) \vee Q(c) \right) \wedge \cdots \wedge \left(\neg P_n(c) \vee Q(c) \right)$$

- Thus, we need to show $\underline{P_1(c) \rightarrow Q(c)}, \underline{P_2(c) \rightarrow Q(c)}, \dots$ are all true

Proof by cases

- **Proof strategy:** Write down all cases and show that implication for all cases are true

$$P_1(c) \rightarrow Q(c) \quad \text{is true}$$

$$P_2(c) \rightarrow Q(c) \quad \text{is true}$$

$$\vdots$$

$$P_n(c) \rightarrow Q(c) \quad \text{is true}$$

- Now, you can use any proof techniques to prove each case

Example

For any real number x , prove that $x \leq |x|$.

Case 1: $x \geq 0$ (positive $\mathbb{R}$) $\mathbb{R}^+$

By def. if $x \geq 0$,

$$\text{then } \underline{|x| = x}$$

then,

$$x \leq |x|$$



$$\begin{cases} x = -2 \\ |x| = 2 = -(-2) \\ -2 \leq -(-2) \end{cases}$$

Case 2: $x < 0$

By def, if $x < 0$

$$\underline{-x = |x|}$$

Since $x < 0$

$$x \leq -x = |x|$$



Example

Prove that $2x^2 + y^2 = 14$ has no integer solution.

Bounds

$$2x^2 \leq 14$$

$$x^2 \leq 7 \leq \underline{9}$$

$$|x| < 3$$

$$x \in \{-2, -1, 0, 1, 2\}$$

$$2x^2, \text{ if } x=0$$

$$x=\pm 1$$

$$x=\pm 2$$

$$2x^2=0$$

$$2x^2=2$$

$$2x^2=8$$

$$\rightarrow y^2 \neq 14 \quad \times$$

$$\rightarrow y^2 = 12 \quad \times$$

$$\rightarrow y^2 = 6 \quad \times$$

$$y^2 \leq 14 < 16$$

$$|y| < 4$$

$$y \in \{-3, -2, -1, 0, 1, 2, 3\}$$

$$y^2$$

